
Work must be shown to receive credit. Each problem is worth 4 pts unless stated otherwise.

1. Use properties of determinants to calculate the determinant of the following matrices:

a. $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 5 & 2 & 1 & 0 \\ 9 & 8 & 9 & 1 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

c. $\begin{bmatrix} 8 & 3 & 5 & 6 \\ 3 & 9 & 1/2 & 7 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 6 & 7 \end{bmatrix}$

2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & 3 \end{bmatrix}$

- a. Calculate the determinant of A using cofactors.

- b. Find C_{32} , i.e., A 's cofactor for the 3rd row and 2nd column.

- c. Does $\sum_{j=1}^3 a_{1j}C_{1j} = \sum_{j=1}^3 a_{3j}C_{3j}$ for any 3×3 matrix?

3. Let $\begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- a. Find x_1 using Cramer's rule.

- b. Find x_2 using Cramer's rule.

4. Suppose that A and B are $n \times n$ matrices such that $\det A = 5$ and $\det B = -2$.

a. Find $\det AB$.

b. Prove that $(AB)^{-1}$ exists.

c. Find $\det (AB)^{-1}$.

5. Find the characteristic polynomials of the following matrices.

a.
$$\begin{bmatrix} 4 & 4 & 1 & 1 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b.
$$\begin{bmatrix} -1 & -2 & 1 \\ 1 & 3 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

6. Find the eigenvalues of the following matrices.

a. Let A be a 3×3 matrix and $p_A(t) = (t+1)(t-2)(t+3)$.

b. Let A be a 2×2 matrix and $p_A(t) = t^2 + 5$.

7. Give a definition for either *eigenvalue* or a *diagonalizable matrix* (or *linear transformation*)

8. [15 points] Let $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$. Find the eigenvalues and eigenvectors for A .

9. Prove the following statement: If A and B are similar matrices then $\det(A) = \det(B)$.

10. [Bonus] Prove that if A' is obtained from the $n \times n$ matrix A by exchanging two rows then $\det A' = -\det A$ (hint: use cofactors and induction).